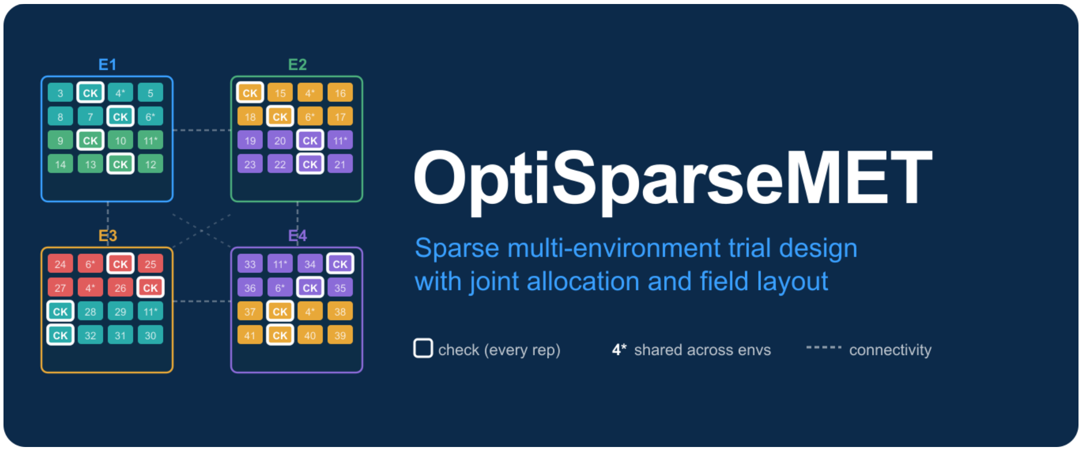

The OptiSparseMET Breeder's Guide

Designing connected, seed-feasible, and statistically robust sparse multi-environment trials

Félicien Akohoue, PhD

Quantitative Genetics

Version 0.2.0 | September 2026

Purpose. This guide explains the decisions made by OptiSparseMET, the evidence required for each decision, and the checks that should be completed before a field book is released. It is written for breeders, quantitative geneticists, trial managers, and reviewers.

Companion materials: package vignette and function reference.

Contents

Contents continued

How to use this guide. Read Sections 1-3 before choosing inputs. Use Sections 4-9 while building the design. Use Sections 10-12 to evaluate and release the fieldbooks. Section 12.5 and Appendices A-B document the September 2026 production additions.

1. The design problem

A multi-environment trial (MET) supports selection across locations, years, or location-by-year combinations. Complete testing is often impossible because candidate numbers exceed local field capacity. Sparse testing therefore observes a planned subset of genotype-by-environment cells.

The central question is not how many cells can be omitted, but which omissions preserve useful information about genetic merit and genotype-by-environment interaction (G×E). A defensible design must also fit the field, respect the available seed, and remain useful if uncertain variance or environmental assumptions prove inaccurate.

1.1 Four linked decisions

1. Define the target population of environments (TPE), the candidate genotypes, and the selection target.
2. Determine which genotypes connect environments and which are allocated sparsely.
3. Assign site-specific replication without exceeding plots or the network-wide seed inventory.
4. Construct local field layouts and compare their prediction precision and selection performance across plausible scenarios.

> **Design principle.** OptiSparseMET treats across-environment allocation and within-environment field layout as statistically coupled. A balanced incidence matrix cannot compensate for an inefficient local layout, and a strong local layout cannot recover genotype-by-environment cells that were never connected.

1.2 Design-stage choices and released-design commitments

OptiSparseMET distinguishes choices explored during planning from commitments enforced after a design is selected. A parameter that is not hard coded is not therefore free to change in a released fieldbook.

- Common-set size and membership can be selected during planning. If the user supplies common_treatments, those identities are fixed immediately.
- After selection, every chosen common genotype is fixed as present in every environment. Re-running the planner may propose a new design; the released design does not change by itself.
- Fixed presence does not require equal replication: a common genotype may have different plot counts at different sites.
- Environmental modalities receive calibrated weights only when valid historical MET responses support them; without that evidence, separate kernels are evaluated without arbitrary modality weights.

- A descriptive environmental group is not automatically treated as a genetic mega-environment.
- Dominance is optional and is not added to non-hybrid material by default.
- Seed is one network inventory: only sites where a genotype is present consume its seed, and amounts spent at one site cannot be spent again.

2. Evidence and inputs

Start from the decision and assemble only the evidence that can be defended. The package can produce a valid structural design from minimal inputs, then use additional genetic, environmental, and operational information when available.

Input class	Minimum	Stronger optional evidence
Trial network	Unique genotype and environment identifiers	TPE definition, site roles, years, target traits
Capacity	Entry or plot capacity by environment	Rows, columns, blocks, checks, p-rep limits, cost
Seed	None if unconstrained	Grams by genotype, grams per plot by site, explicit reserve policy
Genetic structure	None	Family, pedigree A, genomic relationship matrix (GRM), parental markers
Environmental evidence	None	Weather, soil, management, geography, historical years
Historical response	None	Adjusted genotype or G×E values by environment and year
Variance	Positive working values	Historical variance components and uncertainty ranges

2.1 Required identifier discipline

- Use unique, non-missing genotype and environment identifiers.
- Give every matrix matching row and column names.
- Align field-book, seed, pedigree, marker, and environmental identifiers before optimisation.
- Treat each location-by-year combination as a distinct environment when years are analysed explicitly.

2.2 Historical MET responses are helpful, not compulsory

Historical adjusted means, best linear unbiased estimates, or best linear unbiased predictions can calibrate genetic environment covariance. Their use is justified only when the response scale, genotype identities, environments, and years are sufficiently comparable.

If valid historical responses are unavailable, do not invent environmental weights. OptiSparseMET uses the identity matrix as the central genetic environment covariance and carries each environmental kernel as a separate structural sensitivity scenario.

3. Environmental evidence and covariance

Environmental similarity describes sites; genetic environment covariance describes correlated genetic responses across sites. These objects may be related, but they are not interchangeable. The package therefore separates environmental feature construction, descriptive grouping, and genetic covariance calibration.

3.1 Main environmental kernels

Kernel	Representative evidence	Interpretation
Weather	Date-aware temperature, radiation, rainfall, humidity, wind, stress and phenology features	Similarity in atmospheric and crop-stage exposure
Soil	Multi-depth properties, root-zone summaries, uncertainty widths, texture and water availability	Similarity in the edaphic production environment
Management	Planting, irrigation, fertilisation, density, crop protection and other operations	Similarity in planned or delivered management
Geography	Latitude and longitude	Spatial proximity; not a substitute for measured environment

qc_environmental_data() validates ranges, coverage, duplicate keys, missingness, provenance, and imputation. build_environment_kernels() then normalises each modality separately, so a high-dimensional weather block does not dominate because it has more columns.

3.2 Environmental interactions

OptiSparseMET distinguishes: (i) nonlinear similarity within one modality, (ii) omnibus interactions between whole modalities, and (iii) dedicated interactions among named variables. Weather-by-soil, weather-by-management, and soil-by-management kernels represent broad cross-modality hypotheses. By default, centred functional-analysis-of-variance products reduce duplication of the parent main-effect kernels.

> **Interpretation.** Constructing an interaction kernel does not assert that its genetic effect is non-zero. Conversely, failure to estimate a historical weight is not evidence that the physical interaction is absent. The kernel remains a separate uncertainty scenario.

Two environments similar for main effects are not assumed to be equally similar for all interactions. Interaction similarity is calculated from the joint parent feature spaces and can alter pairwise relationships. Nor does main-effect dissimilarity prove a biologically important interaction.

3.2.1 Implicit, omnibus, and dedicated interactions

A Gaussian weather or soil kernel represents nonlinear functions and interactions among all variables in that modality. However, it neither isolates nor tests a named variable combination. Similarly, an omnibus weather-by-soil kernel contains temperature-by-bulk-density functions when both variables survived quality control, but it cannot attribute performance to that pair.

Use build_variable_interaction_kernels() when a particular physiological or management hypothesis must be guaranteed and audited separately. The recommended input is a long data frame with four required columns: interaction, parent, modality, and variable. Each row identifies one exact post-quality-control variable. Rows sharing an interaction and parent define one multicolumn parent, which supports one-hot-encoded categorical management factors.

For example, temperature_x_density contains two rows: temperature, weather, T2M_mean_reproductive; and density, soil, bulk_density_root. The variable names must match the quality-controlled covariate columns exactly. An absent or non-informative variable stops construction rather than being silently substituted.

3.2.2 Construction and strong heredity

Each parent feature is centred and standardised. The package forms their row-wise tensor product, then residualises the tensor features against the intercept and all parent main effects. A Gaussian bandwidth-ensemble kernel is constructed from the residual features. Residualisation therefore reduces main-effect leakage before kernel construction while preserving positive semidefiniteness.

Strong heredity is the default: every dedicated interaction is retained with its exact parent-variable kernels. This is a model-structure guarantee, not a claim that each parent or interaction has a non-zero historical covariance contribution. The tensor-dimension guard prevents unnoticed categorical expansion, while the minimum residual-information rule rejects an interaction that is almost completely explained by its parents.

3.2.3 Audit, historical evidence, and fallback

The variable-interaction audit records the scope, order, matched modalities and variables, raw and retained tensor dimensions, residual-information fraction, effective kernel rank, realised bandwidths, and hierarchy. The variable ledger records every requested column and its parent kernel. Hence, structural incorporation can be verified directly.

When valid historical multi-environment trial responses exist, assess_variable_interactions() compares each dedicated interaction with the same main-effect model. Leave-year-out validation is preferred; leave-environment-out validation is the fallback. Support requires positive mean prediction improvement, a multiplicity-adjusted one-sided paired sign test below the declared threshold, and a positive covariance contribution. Bootstrap inclusion describes selection stability.

Without valid historical responses, no interaction is labelled supported or unsupported. The central genetic environment covariance remains the identity matrix. Every main, omnibus interaction, and dedicated interaction kernel is retained separately for unweighted maximin evaluation. Therefore, a physical hypothesis can inform robustness analysis without being converted into arbitrary genetic covariance.

3.3 Within-year and across-year variability

historical_envirotype() preserves actual dates, crop stages or thermal time, tails, trends, observed-year counts, and interannual variability. assess_envirotype_stability() constructs year-specific relationships and reports their agreement. consensus_relationship() can then form a robust within-modality consensus without allowing a shared unit diagonal or a longer weather archive to create spurious dominance.

- Within-year variability is represented through date- and stage-specific features.
- Across-year variability is represented through year-specific kernels, stability summaries, and uncertainty candidates.
- Years are nested within their modality when clustering stability is assessed.

4. Historical and no-history decision branches

4.1 Valid historical responses

calibrate_environment_covariance() estimates non-negative modality contributions against historical genetic correlations. Leave-environment-out or leave-year-out validation assesses transportability; bootstrap samples quantify uncertainty.

The central model compares main-effect-only and main-plus-interaction candidates by paired blocked validation. An unsupported interaction receives zero central weight, but remains in the robust structural scenario set.

4.2 No valid historical responses

When historical responses are absent or unsuitable, the result has status = "no_historical_met", weights = NULL, and an identity central covariance. Main and interaction kernels are not mixed. The design is evaluated separately under each structure and scored by its worst-case, or maximin, performance.

Decision	With valid historical MET	Without valid historical MET
Central Sigma_E	Cross-validated calibrated covariance	Identity matrix
Modality weights	Estimated and validated	None
Interactions	Central contribution requires validation; all retained as scenarios	All retained as separate scenarios
Robust evaluation	Central plus uncertainty structures	Unweighted maximin across separate structures
Interpretation	Genetic covariance supported by response data	Structural sensitivity, not estimated genetic covariance

Why this matters. A no-history design is not weakened by arbitrary user weights. It is made explicitly conservative: recommendations must perform acceptably across the separate environmental structures.

5. Inferring environmental groups

The user is not expected to know the number of groups in advance. infer_mega_environments() evaluates a feasible range and compares hierarchical clustering with k-means on kernel principal coordinates.

5.1 Acceptance criteria

- Adequate average silhouette separation.
- No group smaller than the declared minimum.
- Agreement across clustering algorithms.
- Agreement across environmental modalities, years, and bootstrap relationship draws.
- No implicit weighting caused by repeated matrices from one modality.

K-means uses finite-descent Hartigan relocation from multiple non-empty starts. Every accepted move strictly decreases the within-cluster sum of squares; therefore, each start terminates without an iteration-limit failure.

5.2 Stability statuses and fallback

Status	Meaning	Operational response
stable	Supported by several independent evidence blocks	May inform stratified capacity, allocation, and sensitivity analysis
provisional	Separated but supported by only one evidence block	Use descriptively and retain network-wide robustness
unstable	No candidate partition passes the requirements	Use one unpartitioned network

Environmental groups become genetic mega-environments only when historical genetic responses support the corresponding G×E structure. An unstable grouping is therefore not forced into treatment allocation or common-set selection.

6. Genetic relationships

Relationship matrices allow information to propagate between related genotypes and help distribute genetic diversity across environments. Their use complements, but does not replace, direct connectivity through shared treatments.

6.1 Additive relationships are the default

- Use a pedigree numerator relationship matrix (A) when a reliable pedigree is available.
- Use a genomic relationship matrix (GRM) when marker information is available.
- Use family labels when neither matrix is available but family structure is known.

6.2 Dominance is optional

Dominance should be requested only for hybrid or other applications in which dominance or specific combining ability contributes to the breeding target. build_hybrid_relationship() constructs additive hybrid relationships from parental relationships; an explicitly requested marker-based dominance matrix can then be combined with additive relationships using defensible variance components.

> **Do not add dominance routinely.** For inbred, pure-line, or other non-hybrid candidates, the additive relationship is the appropriate default unless the biological objective states otherwise.

6.3 Quality checks

- Verify names, dimensions, dosage coding, symmetry, and positive semidefiniteness.
- Tune singular or poorly conditioned matrices by documented bending or blending.
- Record whether the matrix was derived from pedigree, candidates, parents, or hybrids.

7. Capacity and common treatments

7.1 Distinguish entries from physical plots

An environment capacity may describe distinct test entries or physical plots. Checks, common-genotype replication, and partial replication consume additional plots. field_plot_accounting(), required_plots(), and test_entry_capacity() make this distinction explicit.

For equal site size k, J genotypes, I environments, and network replication r, the basic identity is $J \times r = I \times k$. With common genotypes removed, the exact equireplicate identity is $J^* \times r = \Sigma_e k^*_e$. Matching totals are necessary; the binary degree sequence must also be realisable.

7.2 The common-set trade-off

In a selected design, a common genotype has fixed presence in every environment. Common genotypes provide direct cross-site comparisons and stabilise environment contrasts, but each one reduces the capacity available for sparse candidates.

Decision	Statistical benefit	Resource cost
Increase common-set size	More direct connectivity and potentially more precise environment contrasts	Fewer slots for non-common candidates; more seed
Increase local plot replication	Greater local precision for selected anchors	More physical plots and seed at that site
Increase genetic diversity of anchors	Better genetic representation; connectivity count unchanged	May exclude familiar but redundant checks

7.3 Choosing common-set size and identities

The package does not impose one universal number across analyses. During planning, suggest_common_treatments() provides a feasible starting set from seed, family and genetic-group representation, and a connectivity target. optimize_common_treatments() compares nested feasible sizes and chooses the global common-set size and identities. It returns binary presence, not a replication plan. Alternatively, a user may supply preselected common_treatments, which are treated as fixed.

Once selected identities are passed to allocation, every one is fixed as present in every environment. Each distinct shared genotype counts once for a pair, regardless of its plot number. Local replication is assigned afterwards from remaining seed for within-site precision; it never increases connectivity.

With environmental covariance supplied, maximin protects the weakest pair by adapting distinct shared sparse treatments under fixed site sizes; CVaR and mean are alternatives.

7.4 Decision checks

- Every selected common genotype is present at every environment.
- Site-specific replication is feasible with the remaining seed and physical plots.
- The common set represents required families or genetic groups.
- The remaining sparse capacity can still cover the target population.
- The Pareto ledger documents the gain in connectivity against the loss of breadth.

8. Across-environment allocation

8.1 Random-balanced allocation

allocation_method = "random_balanced" implements an M3-inspired, coverage-first allocation. It guarantees that every treatment appears at least once, tolerates unequal site sizes, and then balances remaining replication subject to group and seed feasibility.

8.2 Equireplicate allocation

allocation_method = "equireplicate" implements an M4-type network design. It enforces equal distinct-environment incidence for non-common treatments and exact named environment loads when the slot identity and binary degree sequence are feasible. This is not within-site plot replication. Pairwise concurrence may be refined towards an achievable near-balanced regular-graph design.

Use	Random-balanced	Equireplicate
Priority	Complete coverage with flexible balance	Equal distinct-environment coverage

Use	Random-balanced	Equireplicate
Unequal site sizes	Supported	Supported when degree sequence is realisable
Exact environment-incidence equality	Not guaranteed	Guaranteed in strict mode
Fallback	Natural heterogeneous design	Approximate mode is exploratory and must be labelled

8.3 Colmant contribution optimisation

select_environments(method = "optcontrib") implements the contribution form q'Dq used to select a representative subset of environments. The vector q contains non-negative environment contributions and D represents environmental dissimilarity. The criterion rewards coverage of distinct environmental information rather than selecting only central or mutually similar sites.

The q'Dq criterion addresses environment selection. It does not, by itself, solve genotype allocation, common-set membership, local plot replication, field geometry, or seed feasibility. OptiSparseMET therefore uses it at the relevant decision point and retains coupled prediction-error criteria for the remaining design decisions.

8.4 Genetic and environmental structure in allocation

- Family, GRM, or A groups can be distributed across sites to avoid genetic concentration.
- Stable environmental groups may guide stratification; unstable groups fall back to one network.
- Environmental kernels influence robust design scoring, common-set optimisation, capacity selection, and allocation refinement.

9. Network-wide seed accounting

Seed is an inventory constraint, not an independent site-level allowance. For genotype i, feasibility requires available seed to cover the sum of all delivered plots across environments plus any declared reserve.

> **Presence rule.** If genotype i is absent from site e, its seed use at that site is 0 g. If it is present at several sites—including every site when it is common—its seed use is counted at every one of those sites.

9.1 Reserve policy

The default minimum seed reserve is 0 g. A positive reserve should be specified only when the breeding programme has an explicit retention, germination, replanting, or seed-regeneration policy.

9.2 Allocation and replication sequence

1. Debit the seed required for each allocated genotype-site presence.
2. Protect any explicit genotype-level reserve.
3. Use only the remaining inventory for local partial replication or repeated checks.
4. Reconstruct the final seed ledger from the delivered field books.

9.3 Audit fields

Ledger field	Interpretation
SeedAvailable	Starting network inventory for the genotype
SeedAllocated	Total consumption across all delivered genotype-site plots
ReserveRequired	Protected amount; normally 0 g unless explicitly set
SeedRemaining	Inventory after allocation and replication
Overspent	Must be FALSE for every genotype

10. Within-environment field design

After allocation, each environment receives the exact genotype set and site-specific replication requirements assigned to it. The local engine must then control field heterogeneity without altering those network decisions.

10.1 Block-based repeated-check designs

met_prep_famoptg() constructs augmented, partially replicated (p-rep), and randomised-complete-block-type layouts. It is appropriate when checks anchor incomplete blocks and local heterogeneity is primarily controlled through blocking.

10.2 Alpha row-column stream designs

met_alpha_rc_stream() constructs fixed-grid alpha row-column layouts. It is appropriate when the field has known rows and columns and the analysis will model row, column, first-order autoregressive, or two-dimensional first-order autoregressive spatial structure.

10.3 Local design checklist

- The delivered genotype count equals the allocation for that environment.
- User-fixed local replication is honoured before discretionary p-rep allocation.
- Every repeated check and p-rep plot is counted in physical field size and seed.
- Rows, columns, blocks, traversal order, and unused cells match the field map.
- Relationship-aware dispersion is used only when the chosen matrix is valid and aligned.

11. Statistical evaluation and optimisation

11.1 Information-matrix criteria

met_information() combines genotype-site incidence, delivered plot counts, local design efficiency, the genotype relationship matrix G, and the genetic environment covariance ΣE. Environment means are absorbed as fixed effects.

Criterion	Meaning	Preferred direction
Prediction-error variance (PEV)	Posterior uncertainty of the target genetic value	Lower
Mean coefficient of determination (CDmean)	Mean line reliability relative to its target prior variance	Higher
A-optimality	Average variance of estimable treatment contrasts	Lower criterion; higher efficiency
D-optimality	Generalised variance across estimable contrasts	Lower criterion; higher efficiency

Actual replication matters. A genotype with two delivered plots contributes more local information than a genotype with one; a non-zero incidence cell must not be treated automatically as one plot.

11.2 Outcome simulation

simulate_met() generates G×E effects under $\Sigma E \otimes G$, observes only the delivered cells, predicts by genomic best linear unbiased prediction (GBLUP), and reports selection accuracy and genetic gain. Monte Carlo standard errors and 95% intervals quantify simulation uncertainty.

11.3 Optimisation methods

- Random restart compares independently generated valid layouts.
- Simulated annealing accepts controlled temporary losses to escape local optima.
- The genetic algorithm evolves populations of candidate layouts.

- All methods must preserve the declared margins, seed limits, and fixed replication.

11.4 Robust aggregation

robust_scenarios() crosses plausible genetic and residual variances, covariance shrinkage, and alternative ΣE structures. Maximin selects the design with the best worst-case score and requires no scenario probabilities. Lower-tail conditional value at risk (CVaR) averages the worst fraction when a less conservative but still risk-aware criterion is justified.

Recommended no-history rule. Use maximin across the separate environmental main and interaction kernels. Do not assign probabilities or mixture weights without response evidence.

12. Benchmarking, capacity, Pareto frontiers, and release

12.1 Site-capacity selection

suggest_site_capacity() evaluates nested candidate capacities while holding constrained partner sites fixed. Candidate designs use a common Monte Carlo stream and count site-specific replication and check overhead. The recommendation can maximise accuracy within the candidate interval, meet a target, or identify diminishing returns.

12.2 Benchmarking and validation

A high objective value under one covariance and one variance ratio is not sufficient evidence for release. The proposed design should be compared with explicit reference designs and challenged under the uncertainties that can change a breeding decision.

make_met_benchmark_designs() constructs complete testing as an information upper bound, naive random sparse sampling as a weak control, M3 coverage-first allocation, and strict M4 equireplication when its degree sequence is realisable. Add the proposed optimised design to this comparator set. An infeasible M4 is omitted or labelled approximate; it is never reported as exact.

benchmark_met_designs() uses paired Monte Carlo draws, so every design faces the same realised genetic effects and residual errors. It crosses environmental covariance candidates, genetic and residual variances, and complete site-loss patterns. The statistical ledger reports PEV, CDmean, accuracy, gain, selection coincidence, and rank quality. The operational ledger reports coverage, physical plots, check or border overhead, cost, seed, and site-capacity feasibility.

Environmental validation uses simulate_environment_model_benchmark() for known-truth recovery, benchmark_environment_models() for independence, separate kernels, equal-weight single-K, and historical calibration, and benchmark_environment_missingness() for repeated masking and reconstruction. Equal weighting is a benchmark only; it is never the no-history covariance.

summarize_design_stability() reports common-set frequencies and Jaccard agreement, site-capacity dispersion, and allocation-cell frequencies. A slightly less efficient but stable design may be preferable to an unstable nominal optimum.

12.3 Pareto decisions

A Pareto frontier should display one statistical endpoint at a time. plot_pareto_frontier() places cost minimisation on the lower horizontal axis from 0% at the most expensive observed design to 100% at the least expensive observed design. The upper axis reports the corresponding real cost in decreasing order. The vertical axis reports either prediction accuracy, reliability, or genetic gain on a 0--100% scale. Only observed feasible non-dominated designs are joined by straight segments; dominated designs remain visible in grey.

12.4 Pre-release checklist

- The TPE, selection target, candidate set, and site roles are documented.
- Historical responses have passed comparability checks, or the no-history branch is explicit.
- Environmental main and interaction kernels have provenance and quality-control ledgers.
- Mega-environment status is stable, provisional, or unpartitioned—not silently forced.
- Additive relationships are aligned; dominance is included only when biologically justified.
- Common-set size and identities are documented separately from local replication.
- Every sparse genotype is covered as intended and all exact degree conditions pass.
- Every genotype-site absence consumes 0 g; all presences and replications are debited once.
- The final seed ledger has no overspent genotype and respects the declared reserve.
- Physical plot counts include checks, common replication, and p-rep.
- Field geometry and traversal order are deployable at every site.
- PEV, CDmean, accuracy, gain, and uncertainty are evaluated from delivered plot counts.
- The selected design remains acceptable under the declared robust scenarios.
- Complete, naive-random, M3, feasible strict-M4, and proposed designs are compared at clearly stated budgets.
- Average and worst-scenario PEV, CDmean, accuracy, gain, selection coincidence, and rank quality are acceptable.
- Missing environmental covariates and complete loss of vulnerable sites do not create an unacceptable decision reversal.
- Common-set, site-capacity, and allocation stability are reported across the declared perturbations.
- The combined field book and all audit ledgers are archived together.

12.5 Environmental characterization additions

The environmental workflow now separates descriptive structure from validated hard grouping and exposes the evidence behind that decision. The additions below preserve the package's conservative interpretation while making historical weather and SoilGrids analyses easier to audit.

Candidate discovery and validation

- **infer_mega_environments()** keeps strict defaults for hard groups. If validation fails, **membership** remains one unpartitioned network, while the best estimable structure is retained in the **candidate_*** fields.
- **infer_environmental_strata()** returns the descriptive candidate and permits singleton strata by default. These strata are not genetic mega-environments unless historical GxE responses support that interpretation.
- Block-specific agreement and adjusted-Rand diagnostics show whether instability comes from weather years, soil, another modality, or conflict between evidence blocks.

Request-safe weather and soil metadata

- **available_weather_parameters()** returns supported NASA POWER API codes. The catalogue now separates **api_code**, **output_name**, and advisory **default_aggregation**, so retrieval and interpretation use the correct names.
- SoilGrids requests are validated as property-by-depth combinations. Most properties use the six standard depth intervals; organic carbon stock (**ocs**) is available at 0-30 cm. Unsupported combinations stop with an informative error before network retrieval.

Optional within-block redundancy control

build_environment_kernels() retains **redundancy = "none"** as the default and adds correlation filtering, PCA, and whitening as explicit alternatives. The returned audit preserves original covariates and records retained variables or components, correlation groups, loadings, variance explained, and effective rank before and after control.

One audited historical weather and soil workflow

historical_environment_characterization() connects crop-window weather histories, optional SoilGrids profiles, temporal stability, separate modality kernels, weather-soil agreement, descriptive strata, strict validation, and provenance/QC in one result. Without a historical genetic-response target, it uses a weight-free descriptive consensus; with a defensible target, the existing covariance calibration can estimate supported modality contributions.

How to act on the result

- **candidate_k > 1 and hard_groups = FALSE:** use the partition for description and sensitivity analysis, not as a compulsory allocation rule.
- **Stable weather but low cross-modality agreement:** the weather pattern is repeatable, but soil describes a different geometry; inspect the block diagnostics rather than lowering validation thresholds.
- **hard_groups = TRUE:** environmental validation passed. Continue to require historical GxE evidence before calling the groups genetic mega-environments.

Implementation update: September 2026. See the package function reference for arguments, return fields, and reproducible examples.

13. Decision guide

Question	Preferred package action
No historical MET responses?	Identity central ΣE; no modality weights; maximin across separate kernels
Historical responses span years?	Blocked leave-year-out validation and bootstrap uncertainty
Environmental groups are unstable?	One unpartitioned network; do not force hard groups
Common set not preselected?	Optimise size and identities during planning; fix the selected identities at release
Is additional local precision needed?	Assign seed-feasible local replication; connectivity is unchanged
Non-hybrid candidates?	Use additive relationships; omit dominance
Hybrids with parental markers?	Build additive hybrid G; add explicit dominance only if required
Partner sites have fixed capacity?	Vary only the focal site or subset in the capacity sweep
Seed is scarce?	Apply one network ledger and 0 g at absent genotype-site cells
No explicit seed-retention policy?	Use the default 0 g reserve
Exact equal replication required?	Use strict equireplicate allocation after feasibility checks
Heterogeneous capacities dominate?	Use random-balanced allocation or a feasible unequal equireplicate sequence
Large fixed grid with spatial gradients?	Use the alpha row-column stream design
Need a risk-averse choice without scenario probabilities?	Use maximin robust aggregation

14. Glossary

Term	Definition
A	Pedigree numerator relationship matrix.
CDmean	Mean coefficient of determination; mean reliability of the chosen genetic target.
Common genotype	A genotype fixed as present in every environment of the selected design; local replication may differ.

Term	Definition
CVaR	Conditional value at risk; mean score in a specified lower tail.
Environment	A location, year, or location-by-year trial unit.
G×E	Genotype-by-environment interaction.
GBLUP	Genomic best linear unbiased prediction.
GRM	Genomic relationship matrix.
Kernel	A named matrix describing pairwise similarity under one evidence source or interaction hypothesis.
Maximin	Selection of the design with the best worst-case score.
Mega-environment	A genetically meaningful environmental group supported by response evidence; environmental-only groups are descriptive.
MET	Multi-environment trial.
PEV	Prediction-error variance.
p-rep	Partially replicated design.
ΣE	Genetic environment covariance matrix.
TPE	Target population of environments.

15. References

Colmant, A., Pita, F., and Covarrubias-Pazaran, G. (2026). Some objective functions and ideas to optimize experimental designs in artificial selection programs. Crop Science, 66, e70337.

Kirkpatrick, S., Gelatt, C. D., and Vecchi, M. P. (1983). Optimization by Simulated Annealing. Science, 220, 671–680.

Montesinos-López, O. A., Mosqueda-González, B. A., Salinas-Ruiz, J., Montesinos-López, A., and Crossa, J. (2023). Sparse multi-trait genomic prediction under balanced incomplete block design. The Plant Genome, 16, e20305.

Rincent, R., Laloë, D., Nicolas, S., et al. (2012). Maximizing the reliability of genomic selection by optimizing the calibration set of reference individuals. Genetics, 192, 715–728.

16. Final perspective

A strong sparse MET is not defined by sparsity alone. It is defined by the information retained, the uncertainty acknowledged, the evidence obtained against explicit reference designs, and the field plan that can actually be planted. OptiSparseMET makes those obligations auditable. The breeder still decides which uncertainty is scientifically credible and which operational compromise is acceptable. Release the design only when its statistical evidence, environmental uncertainty, seed ledger, physical plot accounting, and field geometry agree with the breeding objective.

Appendix A Production engine additions

The final design calculation can now use the breeding programme's historical response covariance, target-environment priorities, local field geometry, residual precision, and physical costs in one objective.

Historical genetic covariance

fit_historical_met() fits diagonal, factor-analytic, or unstructured environment covariance by REML. Missing genotype-by-environment cells remain in the likelihood. Supply residual variance when cells are unreplicated; otherwise replicated cells estimate it. Compare convergence, AIC, BIC, blocked prediction, and biological plausibility before selecting the central covariance.

Programme target and site precision

- **tpe_weights** records the relative importance of environments in the target population and is normalised to sum to one.
- **sigma_e2** may differ by environment. Do not use a common value when historical trial analyses support material precision differences.
- TPE weights are programme priorities. They are not environmental-kernel weights and are not inferred from the diagonal of the covariance.

Prediction-optimal allocation

M3 and M4 remain transparent constructors. **allocation_method = prediction_optimal** refines a feasible start against mean PEV, CDmean, or expected gain. With **common_set = optimize_jointly**, common-set size and identities change in the same search as the remaining incidence matrix. **fieldbook_design_evaluator()** can score integer replication, checks, cost, and full local treatment information from plantable layouts.

Search and large-network calculation

Choose exchange, simulated annealing, or genetic mutation-selection independently from the scientific criterion. Guarded exact binary search certifies only small problems and refuses larger ones. **met_information(solver = "auto")** switches to matrix-free PCG for large networks. Record search and solver diagnostics and confirm that rankings are stable across restarts, probes, and tolerances.

Appendix B Release checks and design record

The released fieldbook must reproduce the design that was scored. The package now treats allocation, realised plot counts, covariance assumptions, diagnostics, and provenance as one versioned record.

Integer replication and physical cost

recommend_replication() converts fractional p-rep targets to balanced integer plots. A target of 1.5 gives one plot to half of the occupied cells and two to the remainder. **design_objective()** accepts environment-specific cost per plot and fixed overhead such as repeated checks. Both candidate and overhead plots count against the budget.

Optimisation evidence

- Review proposed, accepted, and improving moves and the acceptance rate.
- Investigate infeasible proposals and fieldbook-evaluator failures.
- Review unique candidate evaluations, cache hits, and every restart trajectory; retain the criterion, search engine, random seed, and controls.

Robust and adaptive decisions

robust_prediction evaluates covariance and variance uncertainty together with TPE shifts, emergence or efficiency loss, plot-cost changes, and complete site loss. **adaptive_met_allocation()** adds the next most informative cells to observations already collected or committed and returns a stepwise marginal-information audit.

Validated release object

sparse_met_design() stores the treatment-by-environment allocation, integer replication, fieldbooks, G, Sigma_E, TPE weights, variance assumptions, diagnostics, and provenance. **validate_sparse_met_design()** checks identifier alignment, covariance symmetry and positive semidefiniteness, and exact agreement between each fieldbook and its replication matrix.

Release checklist

- Every allocated treatment appears in the correct environment and at the exact recorded integer replication.
- Check plots and other fixed overhead are included in capacity and cost.
- The network-wide seed ledger remains non-negative after the reserve.
- Dense or PCG information diagnostics meet the declared numerical rule.
- Robustness and benchmarking cover covariance, variance, missing data, site loss, and explicit reference designs.
- The validated design object, fieldbooks, seed, software version, and analysis assumptions are archived together.